



FATE MONTHLY REPORT

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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
April 2024



PO102958/ESO/20/95952/FLAB

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1 NIGHT1: Statistical analysis of forecast performances

1.1 NIGHT1: Scatter plots (bias, RMSE, SD) for long and short forecast time scale

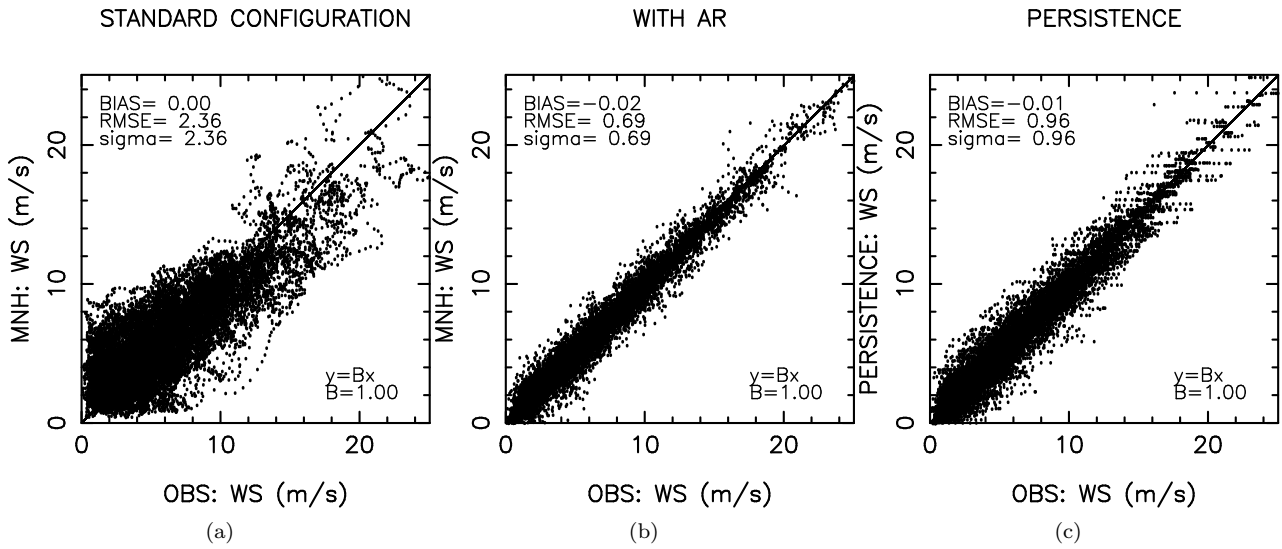


Figure 1: night1 - wind speed (ms^{-1}): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

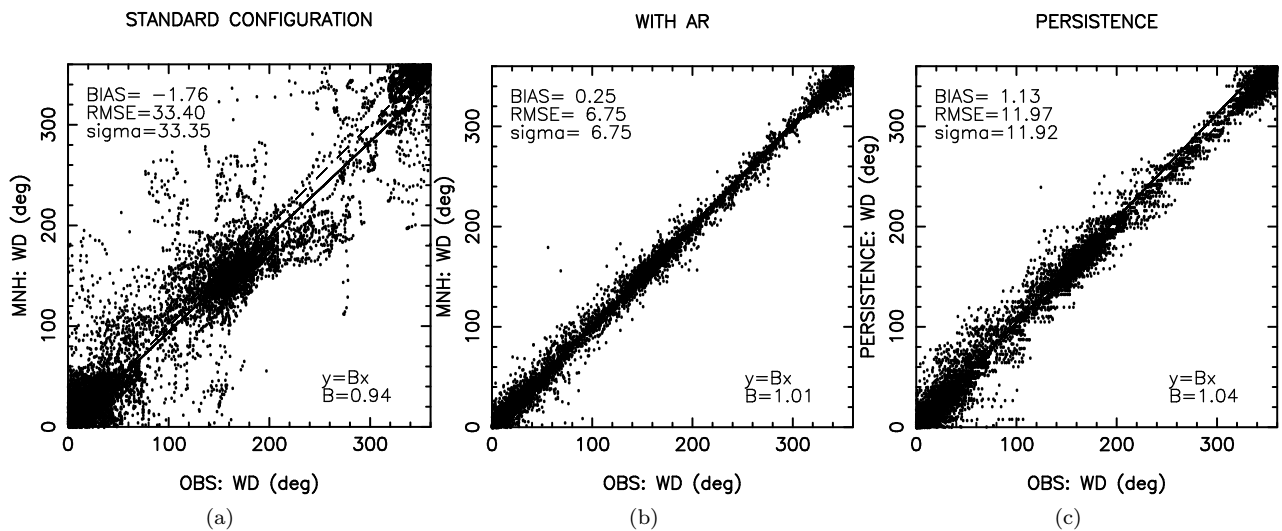


Figure 2: night1 - wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

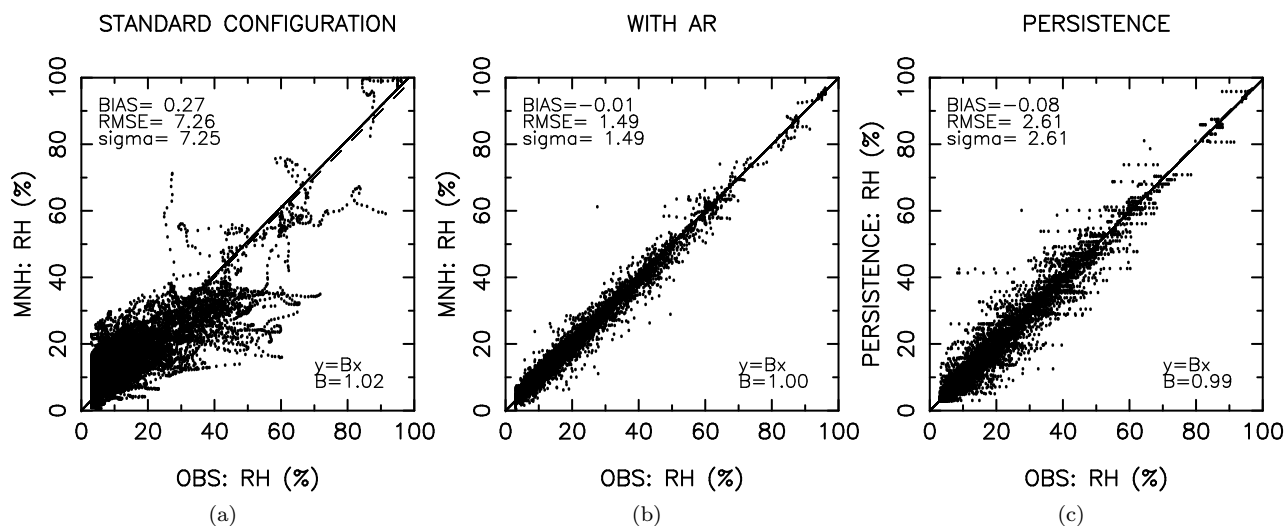


Figure 3: night1 - relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

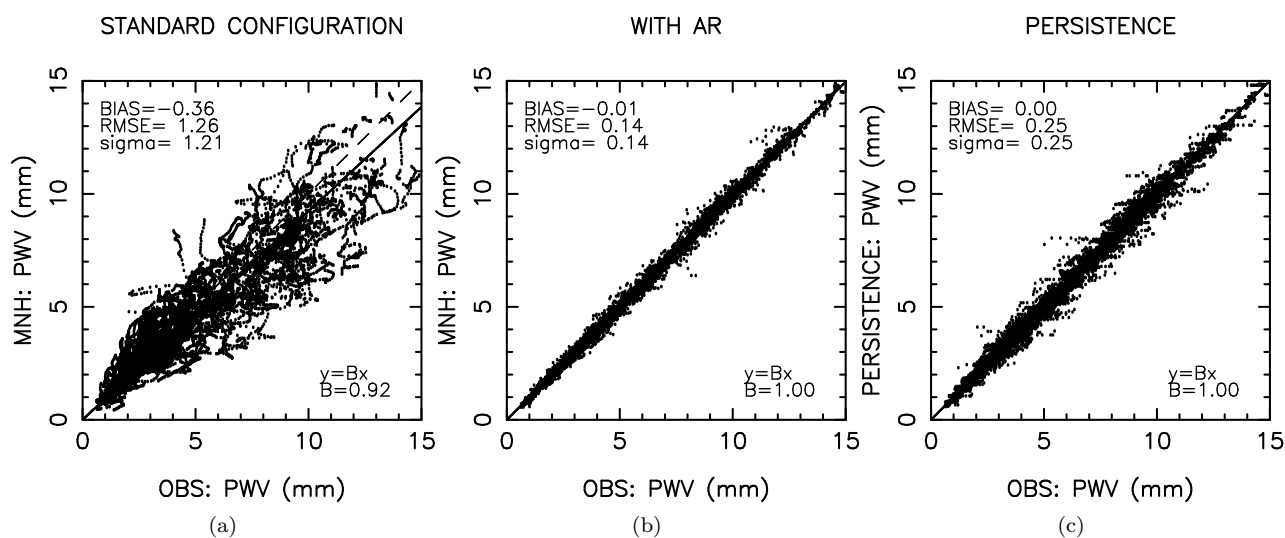


Figure 4: night1 - precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

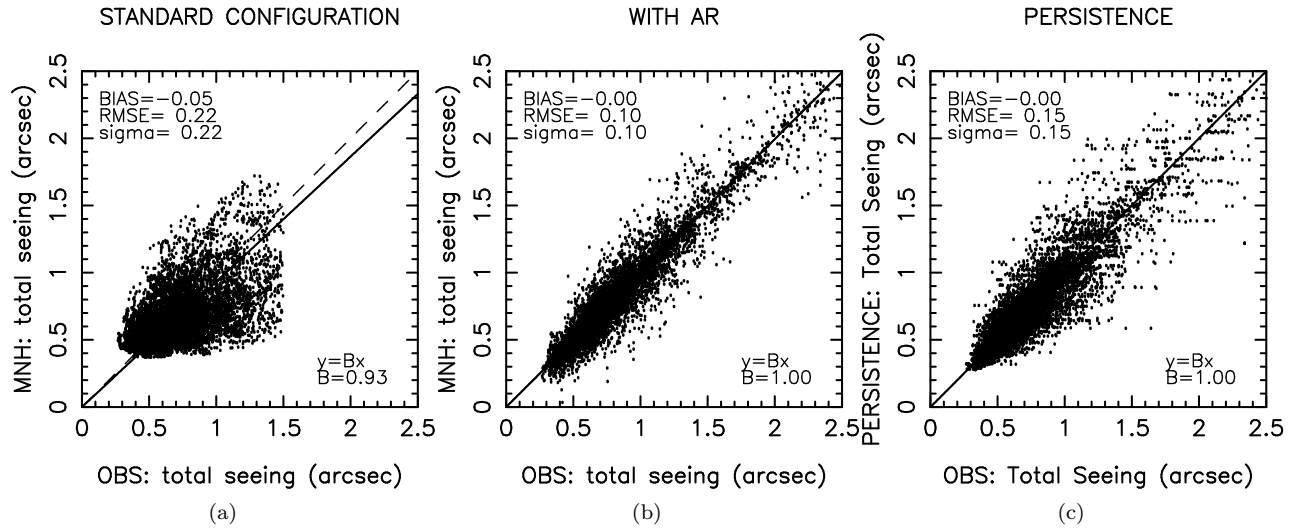
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Figure 5: night1 - total seeing (arcsec): (a) STANDARD CONFIGURATION (< 1.5"), (b) WITH AR (1H), (c) PERSISTENCE (1H)

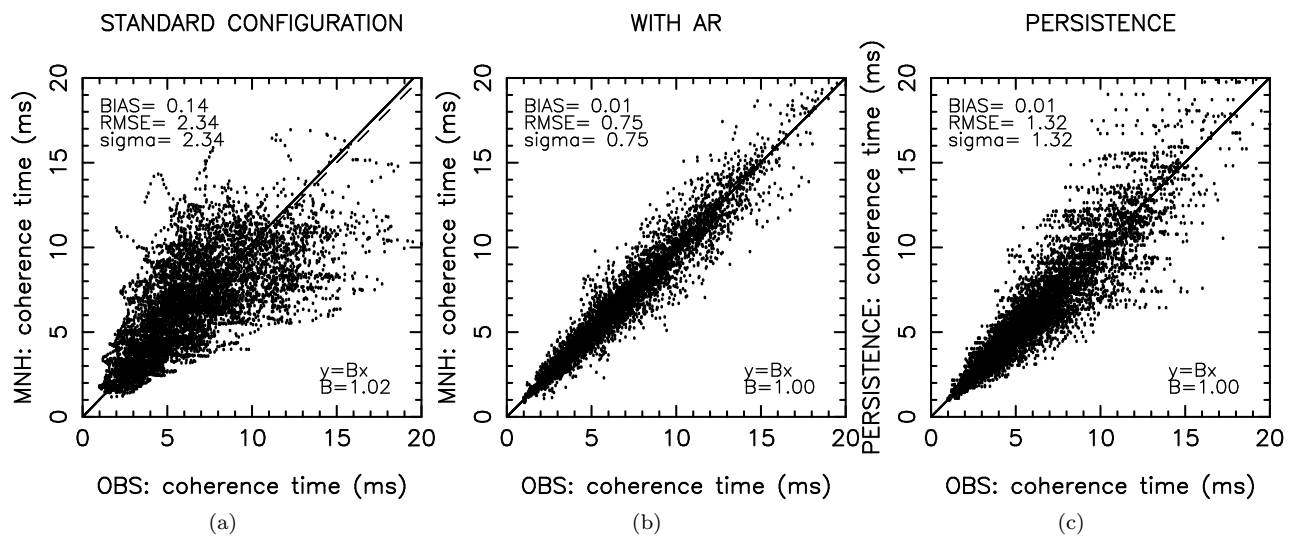


Figure 6: night1 - wavefront coherence time (ms): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

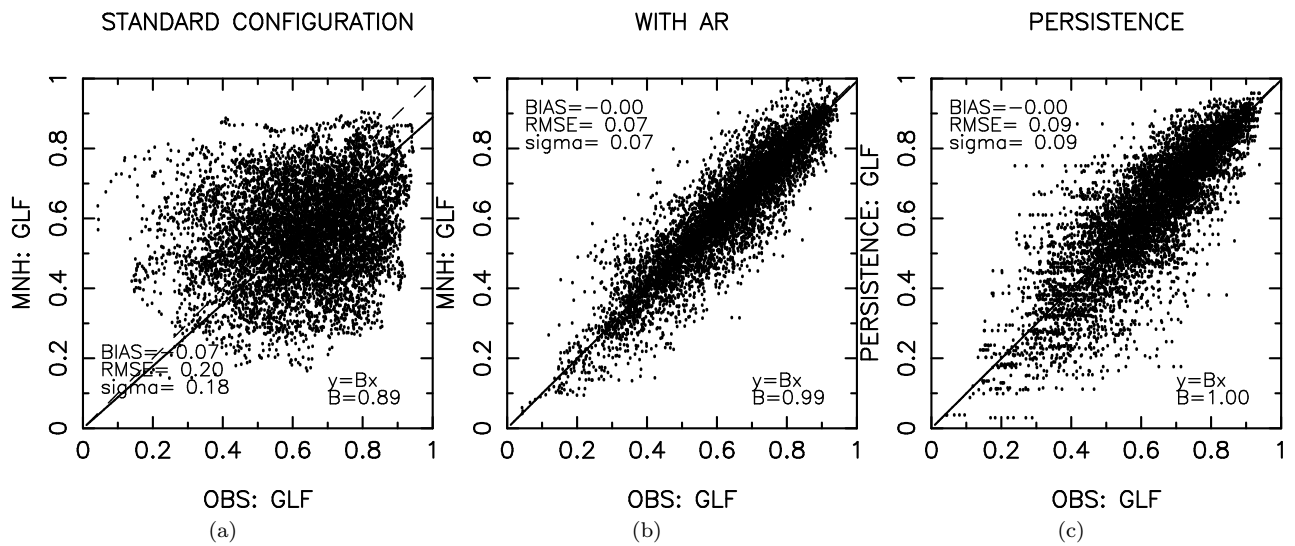


Figure 7: night1 - ground layer fraction (dimensionless): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

1.2 NIGHT1: Summarizing values of bias, RMSE, SD for long and short forecast time scales

VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS ($m \cdot s^{-1}$)	0.00	2.36	2.36	0.18	2.01	2.00
WD (degree)	-1.76	33.40	33.35	-7.26	26.73	25.73
RH (percent)	0.27	7.26	7.25	0.18	7.64	7.64
PWV (mm)	-0.36	1.26	1.21	-0.34	1.24	1.19
SEE (arcsec)	-0.05	0.22	0.22	-0.00	0.18	0.18
TAU (ms)	0.14	2.34	2.34	-0.42	2.23	2.19
GLF (dimensionless)	-0.07	0.20	0.18	-0.10	0.19	0.17

Table 1: night1 - Statistics for variables in standard configuration: incremental month (i.e., since the beginning of service) *vs* last month (April-2024)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	0.01	0.66	0.66		0.86	0.86
WD (degree)	0.06	5.45	5.45		9.67	9.65
RH (percent)	-0.03	1.31	1.31		2.40	2.39
PWV (mm)	-0.01	0.12	0.12		0.23	0.23
SEE (arcsec)	0.00	0.07	0.07		0.10	0.10
TAU (ms)	0.01	0.67	0.67		1.32	1.32
GLF (dimensionless)	-0.00	0.05	0.05		0.08	0.08

Table 2: night1 - Last month (April-2024): statistics for variables with AR (1H) *vs* persistence (1H)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	-0.02	0.69	0.69		0.96	0.96
WD (degree)	0.25	6.75	6.75		11.97	11.92
RH (percent)	-0.01	1.49	1.49		2.61	2.61
PWV (mm)	-0.01	0.14	0.14		0.25	0.25
SEE (arcsec)	-0.00	0.10	0.10		0.15	0.15
TAU (ms)	0.01	0.75	0.75		1.32	1.32
GLF (dimensionless)	-0.00	0.07	0.07		0.09	0.09

Table 3: night1 - Incremental month (i.e., since the beginning of service): statistics for variables with AR (1H) *vs* persistence (1H)

PARAMETER	STANDARD	WITH AR (1H)
POD1	61.8	93.0
POD2	51.2	86.5
POD3	79.2	95.5
PC	64.3	91.8
EBD	1.9	0.0

Table 4: night1 - PODs for wind speed (ms^{-1})

PARAMETER	STANDARD	WITH AR (1H)
POD1	99.5	99.4
POD2	44.1	88.4
POD3	31.4	76.7
PC	93.5	98.1
EBD	0.0	0.0

Table 5: night1 - PODs for wind speed (ms^{-1}) and thresholds (12.0,18.0) ms^{-1}

1.3 NIGHT1: Probability of detection (PODi) for long and short forecast time scales

PARAMETER	STANDARD	WITH AR (1H)
POD1	88.1	98.5
POD2	52.1	86.9
POD3	85.1	98.5
POD4	35.5	92.1
PC	80.4	96.8
EBD	1.4	0.0

Table 6: night1 - PODs for wind direction (degree)

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PARAMETER	STANDARD	WITH AR (1H)
POD1	43.7	90.9
POD2	62.4	93.2
POD3	83.5	96.6
PC	64.3	93.7
EBD	3.1	0.0

Table 7: night1 - PODs for relative humidity (percent)

PARAMETER	STANDARD	WITH AR (1H)
POD1	99.7	99.8
POD2	36.5	88.0
POD3	77.6	97.4
PC	98.1	99.5
EBD	0.0	0.0

Table 8: night1 - PODs for relative humidity (percent) and thresholds (50,70) percent

PARAMETER	STANDARD	WITH AR (1H)
POD1	82.4	97.7
POD2	72.2	95.9
POD3	88.9	99.1
PC	82.8	97.9
EBD	0.1	0.0

Table 9: night1 - PODs for precipitable water vapor (mm)

PARAMETER	STANDARD	WITH AR (1H)
POD1	70.8	89.4
POD2	37.0	72.1
POD3	42.9	89.4
PC	51.2	83.8
EBD	6.6	0.4

Table 10: night1 - PODs for total seeing (arcsec) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	96.2	99.8
POD2	97.0	99.3
POD3	79.0	98.5
PC	94.1	99.5
EBD	4.9	0.3

Table 11: night1 - PODs for total seeing (arcsec) and accuray 0.24

PARAMETER	STANDARD	WITH AR (1H)
POD1	65.7	93.2
POD2	44.4	85.5
POD3	86.8	94.8
PC	68.9	91.5
EBD	1.1	0.0

Table 12: night1 - PODs for wavefront coherence time (ms) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	87.2	99.9
POD2	77.0	98.6
POD3	91.5	99.5
PC	87.9	99.4
EBD	3.1	0.1

Table 13: night1 - PODs for wavefront coherence time (ms) and accuray 1.22

PARAMETER	STANDARD	WITH AR (1H)
POD1	58.2	87.2
POD2	38.3	70.7
POD3	21.1	84.3
PC	38.0	80.5
EBD	17.2	0.4

Table 14: night1 - PODs for ground layer fraction (dimensionless) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	87.6	99.4
POD2	85.8	98.9
POD3	44.2	97.9
PC	76.5	99.1
EBD	18.3	0.5

Table 15: night1 - PODs for ground layer fraction (dimensionless) and accuray 0.14

2 DAY1: Statistical analysis of forecast performances related

2.1 DAY1: Scatter plots (bias, RMSE, SD) for long and short forecast time scale

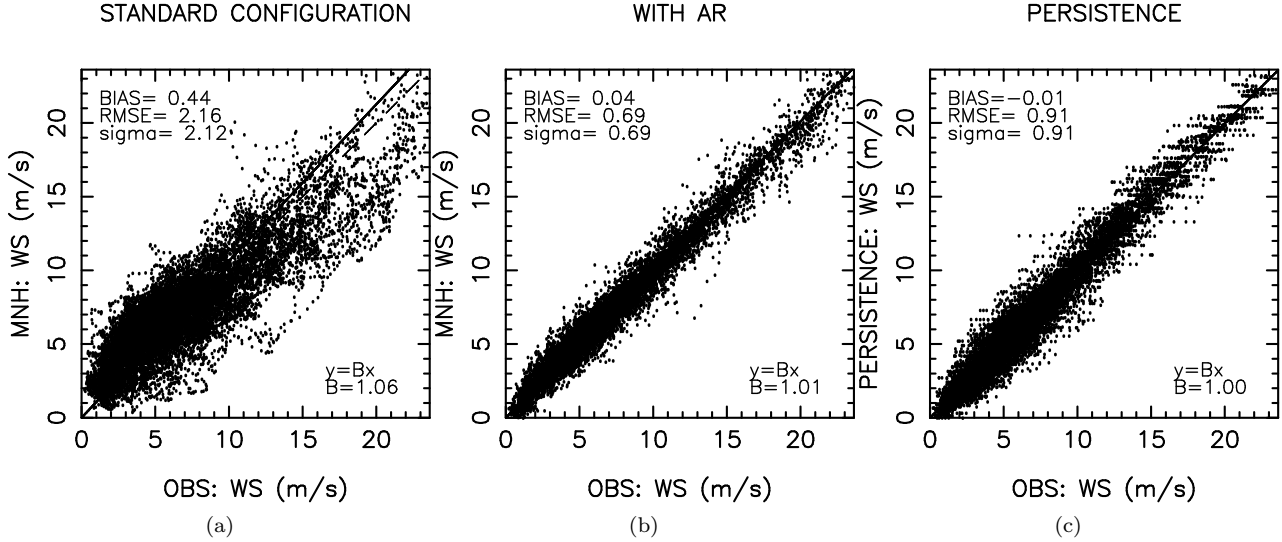


Figure 8: day1 - wind speed (ms^{-1}): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

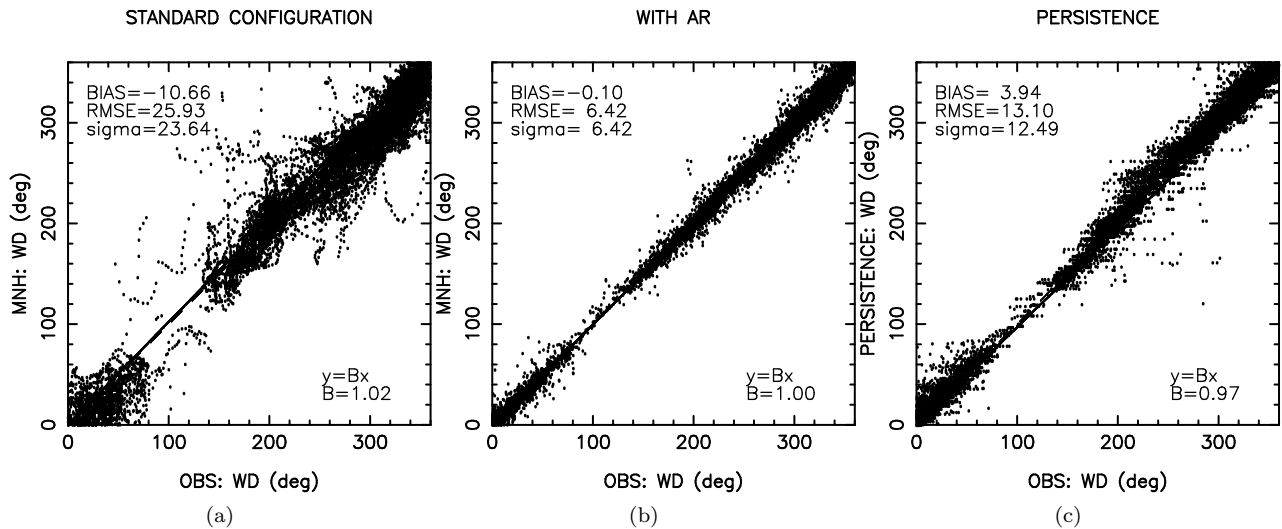


Figure 9: day1 - wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

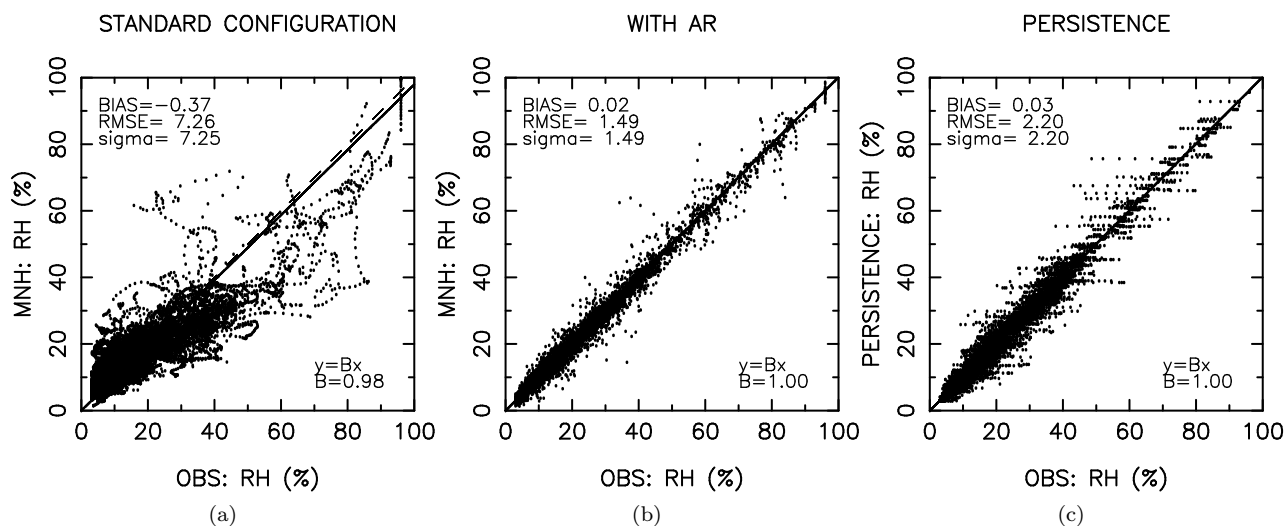


Figure 10: day1 - relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

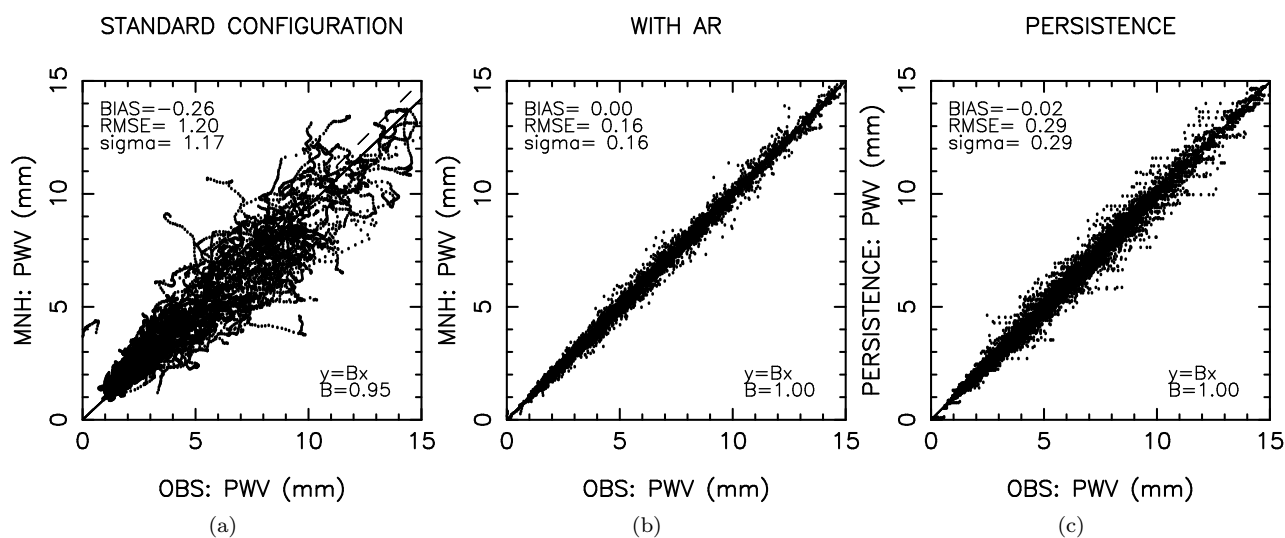


Figure 11: day1 - precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

2.2 DAY1: Summarizing values of bias, RMSE, SD for long and short forecast time scales

VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS ($m \cdot s^{-1}$)	0.44	2.16	2.12	0.55	2.07	1.99
WD (degree)	-10.66	25.93	23.64	-8.76	22.40	20.62
RH (percent)	-0.37	7.26	7.25	-1.52	7.72	7.57
PWV (mm)	-0.26	1.20	1.17	-0.17	1.30	1.29

Table 16: day1 - Statistics for variables in standard configuration: incremental month (i.e., since the begining of service) *vs* last month (April-2024)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	-0.01	0.72	0.72		0.86	0.86
WD (degree)	-0.29	6.32	6.31		14.23	13.82
RH (percent)	-0.01	1.61	1.61		2.56	2.56
PWV (mm)	-0.00	0.15	0.15		0.29	0.28

Table 17: day1 - Last month (April-2024): statistics for variables with AR (1H) *vs* persistence (1H)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	0.04	0.69	0.69		0.91	0.91
WD (degree)	-0.10	6.42	6.42		13.10	12.49
RH (percent)	0.02	1.49	1.49		2.20	2.20
PWV (mm)	0.00	0.16	0.16		0.29	0.29

Table 18: day1 - Incremental month (i.e., since the beginning of service): statistics for variables with AR (1H) *vs* persistence (1H)

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PARAMETER	STANDARD	WITH AR (1H)
POD1	43.3	88.0
POD2	65.9	89.7
POD3	81.2	93.5
PC	65.1	90.5
EBD	1.1	0.0

Table 19: day1 - PODs for wind speed (ms^{-1})

PARAMETER	STANDARD	WITH AR (1H)
POD1	98.1	99.2
POD2	52.3	87.2
POD3	33.3	85.1
PC	92.2	97.7
EBD	0.2	0.0

Table 20: day1 - PODs for wind speed (ms^{-1}) and thresholds (12.0,18.0) ms^{-1} **2.3 DAY1: Probability of detection (PODi) for long and short forecast time scales**

FATE - Monthly Report

PARAMETER	STANDARD	WITH AR (1H)
POD1	67.5	94.8
POD2	37.2	88.6
POD3	85.6	97.4
POD4	89.7	94.4
PC	76.2	94.8
EBD	0.3	0.0

Table 21: day1 - PODs for wind direction (degree)

PARAMETER	STANDARD	WITH AR (1H)
POD1	29.7	88.6
POD2	66.4	94.3
POD3	89.4	97.2
PC	70.9	94.7
EBD	1.4	0.0

Table 22: day1 - PODs for relative humidity (percent)

PARAMETER	STANDARD	WITH AR (1H)
POD1	99.6	99.8
POD2	27.2	90.7
POD3	39.3	92.0
PC	97.3	99.5
EBD	0.3	0.0

Table 23: day1 - PODs for relative humidity (percent) and thresholds (50,70) percent

PARAMETER	STANDARD	WITH AR (1H)
POD1	66.8	95.9
POD2	75.4	96.7
POD3	91.8	98.8
PC	83.2	97.7
EBD	0.3	0.0

Table 24: day1 - PODs for precipitable water vapor (mm)

3 NIGHT2: Statistical analysis of forecast performances

3.1 NIGHT2: Scatter plots (bias, RMSE, SD) for long and short forecast time scale

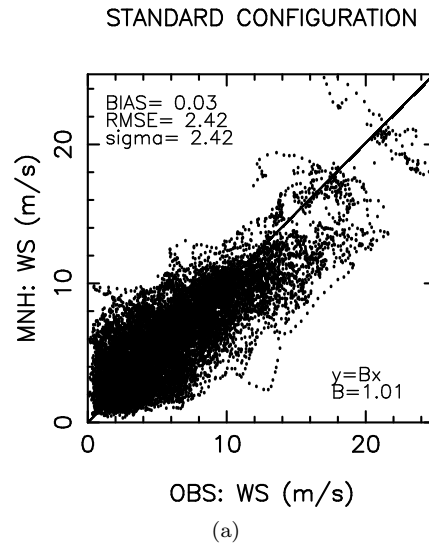


Figure 12: night2 - wind speed (ms^{-1}): STANDARD CONFIGURATION

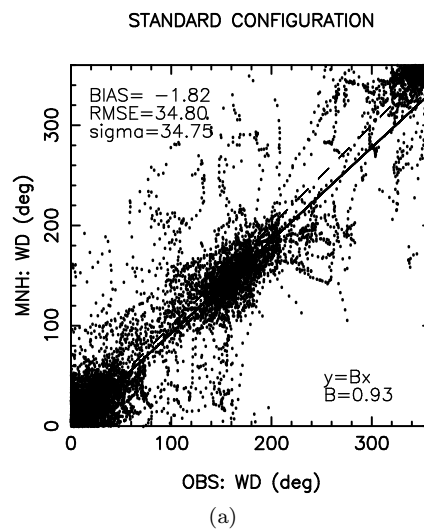


Figure 13: night2 - wind direction (degree): STANDARD CONFIGURATION

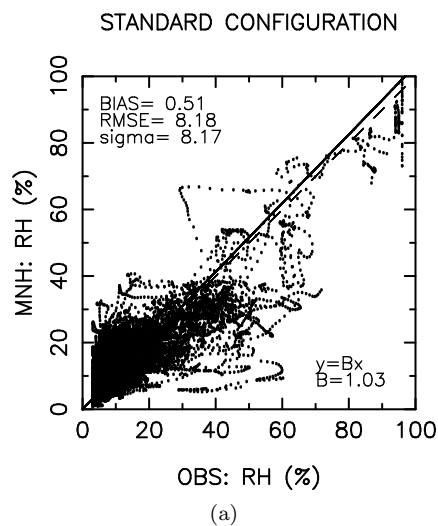


Figure 14: night2 - relative humidity (percent): STANDARD CONFIGURATION

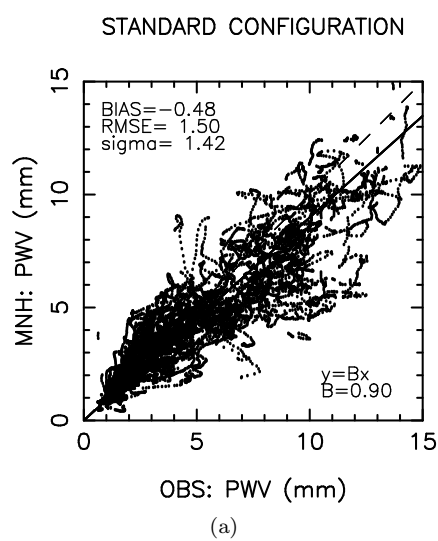


Figure 15: night2 - precipitable water vapor (mm): STANDARD CONFIGURATION

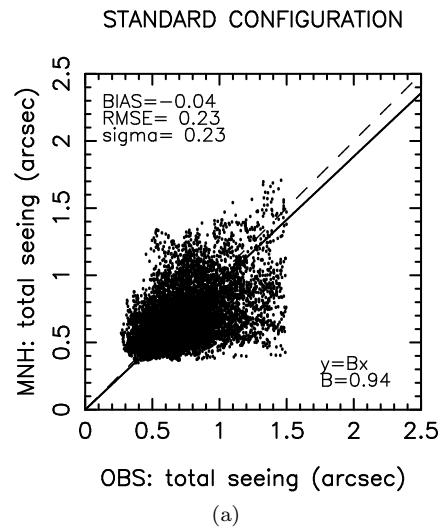


Figure 16: night2 - total seeing (arcsec): STANDARD CONFIGURATION ($< 1.5''$)

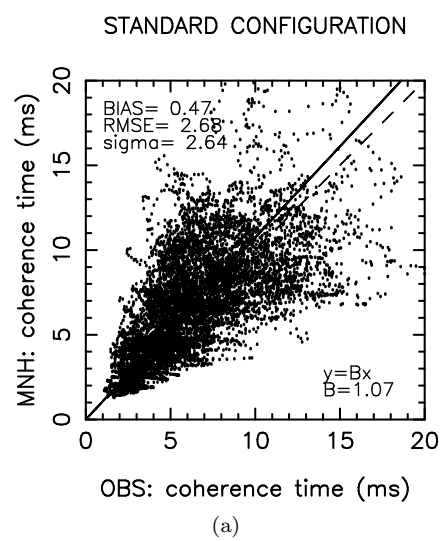


Figure 17: night2 - wavefront coherence time (ms): STANDARD CONFIGURATION

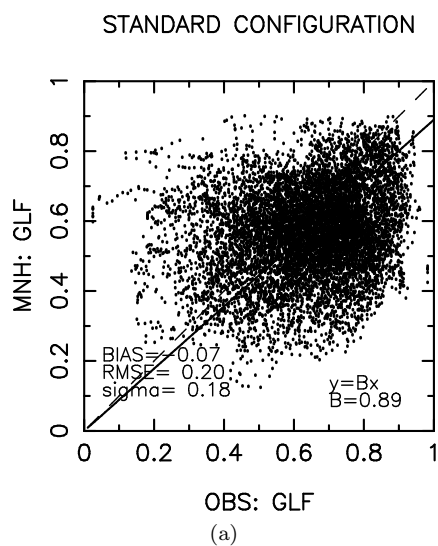


Figure 18: night2 - ground layer fraction (dimensionless): STANDARD CONFIGURATION

3.2 NIGHT2: Summarizing values of bias, RMSE, SD for long and short forecast time scales

VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS (m·s ⁻¹)	0.03	2.42	2.42	0.09	2.08	2.08
WD (degree)	-1.82	34.80	34.75	-5.95	25.97	25.28
RH (percent)	0.51	8.18	8.17	-1.08	8.45	8.38
PWV (mm)	-0.48	1.50	1.42	-0.28	1.34	1.31
SEE (arcsec)	-0.04	0.23	0.23	-0.03	0.21	0.21
TAU (ms)	0.47	2.68	2.64	0.38	2.44	2.41
GLF (dimensionless)	-0.07	0.20	0.18	-0.08	0.19	0.17

Table 25: night2 - Statistics for variables in standard configuration: incremental month (i.e., since the beginning of service) *vs* last month (April-2024)

PARAMETER	STANDARD
POD1	59.3
POD2	52.1
POD3	79.0
PC	63.5
EBD	2.2

Table 26: night2 - PODs for wind speed (ms^{-1})

3.3 NIGHT2: Probability of detection (PODi) for long and short forecast time scale

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PARAMETER	STANDARD
POD1	89.2
POD2	49.2
POD3	78.8
POD4	24.4
PC	78.3
EBD	1.6

Table 27: night2 - PODs for wind direction (degree)

PARAMETER	STANDARD
POD1	38.2
POD2	57.4
POD3	81.0
PC	60.2
EBD	4.3

Table 28: night2 - PODs for relative humidity (percent)

PARAMETER	STANDARD
POD1	81.3
POD2	64.2
POD3	89.1
PC	80.6
EBD	0.4

Table 29: night2 - PODs for precipitable water vapor (mm)

PARAMETER	STANDARD
POD1	67.2
POD2	37.9
POD3	44.4
PC	50.6
EBD	8.0

Table 30: night2 - PODs for total seeing (arcsec) and accuray 0.0

PARAMETER	STANDARD
POD1	95.1
POD2	95.4
POD3	77.4
PC	92.8
EBD	5.6

Table 31: night2 - PODs for total seeing (arcsec) and accuray 0.24

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PARAMETER	STANDARD
POD1	62.7
POD2	42.7
POD3	88.8
PC	69.1
EBD	1.3

Table 32: night2 - PODs for wavefront coherence time (ms) and accuray 0.0

PARAMETER	STANDARD
POD1	85.6
POD2	67.7
POD3	91.3
PC	84.7
EBD	3.3

Table 33: night2 - PODs for wavefront coherence time (ms) and accuray 1.22

PARAMETER	STANDARD
POD1	54.1
POD2	40.0
POD3	20.9
PC	36.8
EBD	16.6

Table 34: night2 - PODs for ground layer fraction (dimensionless) and accuray 0.0

PARAMETER	STANDARD
POD1	89.6
POD2	87.3
POD3	44.3
PC	77.3
EBD	18.0

Table 35: night2 - PODs for ground layer fraction (dimensionless) and accuray 0.14

4 DAY2: Statistical analysis of forecast performances

4.1 DAY2: Scatter plots (bias, RMSE, SD) for long and short forecast time scale

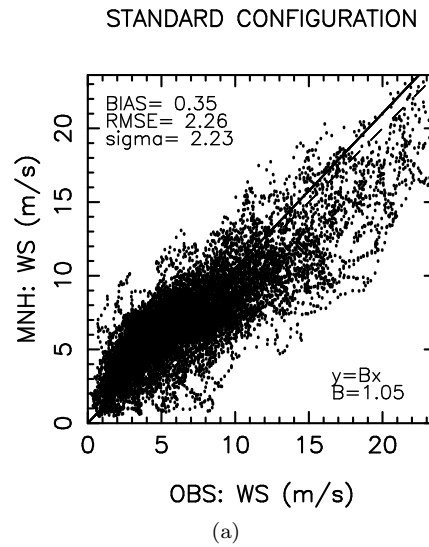


Figure 19: day2 - wind speed (ms^{-1}): STANDARD CONFIGURATION

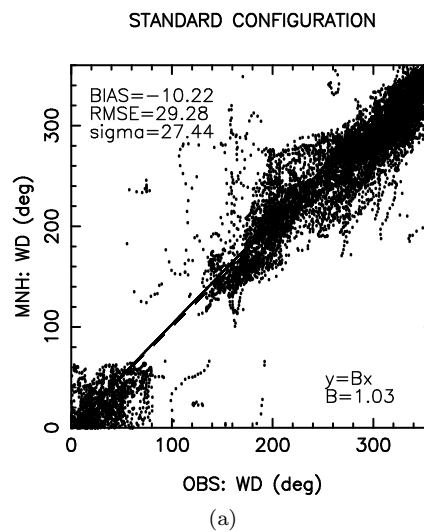


Figure 20: day2 - wind direction (degree): STANDARD CONFIGURATION

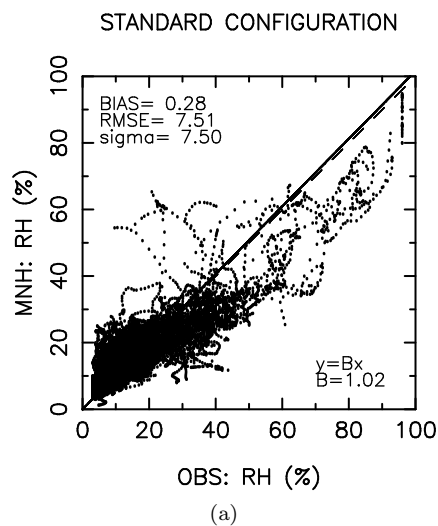


Figure 21: day2 - relative humidity (percent): STANDARD CONFIGURATION

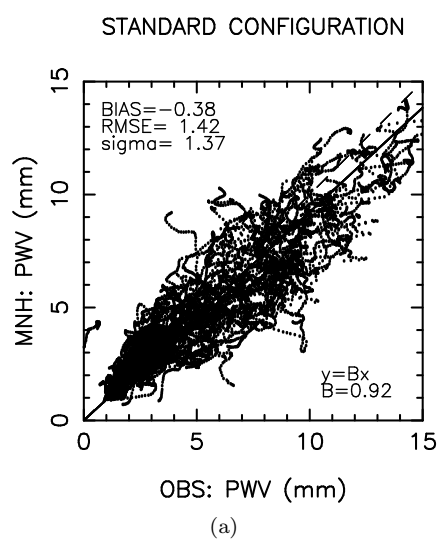


Figure 22: day2 - precipitable water vapor (mm): STANDARD CONFIGURATION

4.2 DAY2: Summarizing values of bias, RMSE, SD for long and short forecast time scales

VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS (m·s ⁻¹)	0.35	2.26	2.23	0.57	2.02	1.93
WD (degree)	-10.22	29.28	27.44	-10.41	23.35	20.90
RH (percent)	0.28	7.51	7.50	-1.03	6.64	6.56
PWV (mm)	-0.38	1.42	1.37	-0.17	1.49	1.48

Table 36: day2 - Statistics for variables in standard configuration: incremental month (i.e., since the begining of service) *vs* last month (April-2024)

FATE - Monthly Report

PARAMETER	STANDARD
POD1	43.8
POD2	66.1
POD3	77.2
PC	63.9
EBD	1.0

Table 37: day2 - PODs for wind speed (ms^{-1})**4.3 DAY2: Probability of detection (PODi) for long and short forecast time scales**

FATE - Monthly Report

PARAMETER	STANDARD
POD1	64.4
POD2	29.6
POD3	77.7
POD4	89.3
PC	73.1
EBD	0.5

Table 38: day2 - PODs for wind direction (degree)

PARAMETER	STANDARD
POD1	22.6
POD2	57.6
POD3	88.2
PC	65.4
EBD	1.9

Table 39: day2 - PODs for relative humidity (percent)

PARAMETER	STANDARD
POD1	66.0
POD2	66.9
POD3	89.6
PC	79.3
EBD	0.3

Table 40: day2 - PODs for precipitable water vapor (mm)

5 NIGHT3: Statistical analysis of forecast performances

5.1 NIGHT3: Scatter plots (bias, RMSE, SD) for long and short forecast time scale

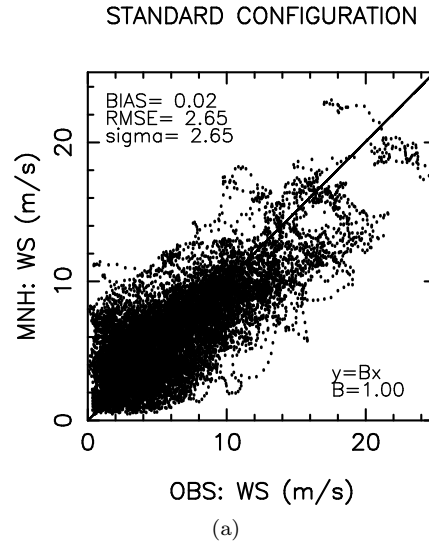


Figure 23: night3 - wind speed ($m s^{-1}$): STANDARD CONFIGURATION

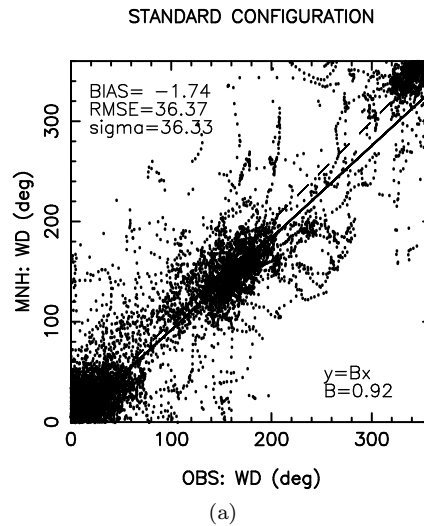


Figure 24: night3 - wind direction (degree): STANDARD CONFIGURATION

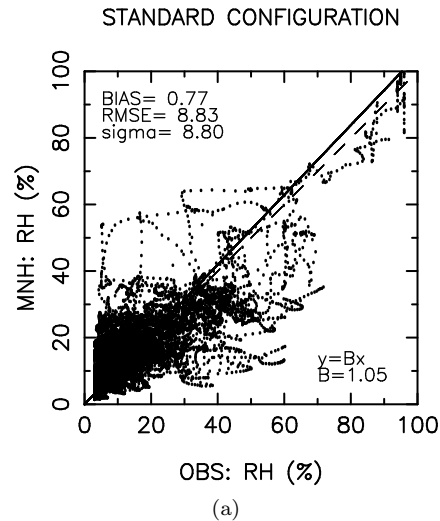


Figure 25: night3 - relative humidity (percent): STANDARD CONFIGURATION

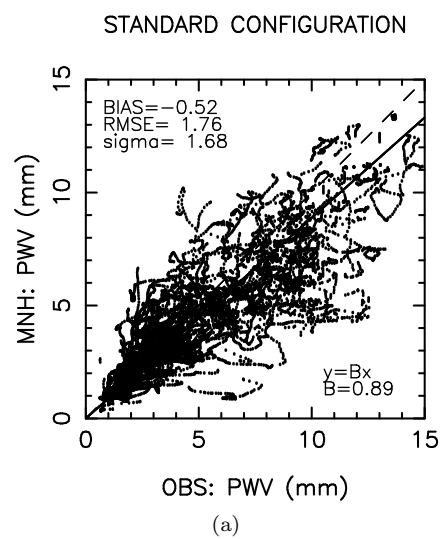


Figure 26: night3 - precipitable water vapor (mm): STANDARD CONFIGURATION

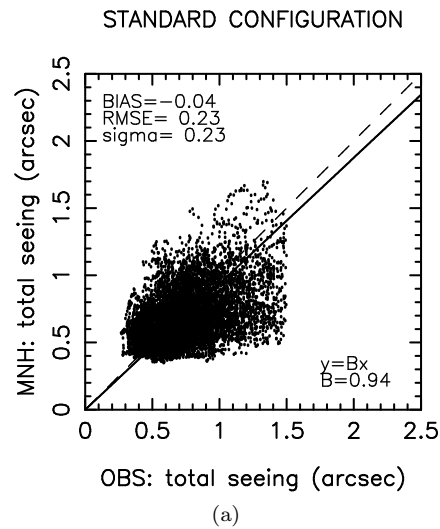


Figure 27: night3 - total seeing (arcsec): STANDARD CONFIGURATION (< 1.5")

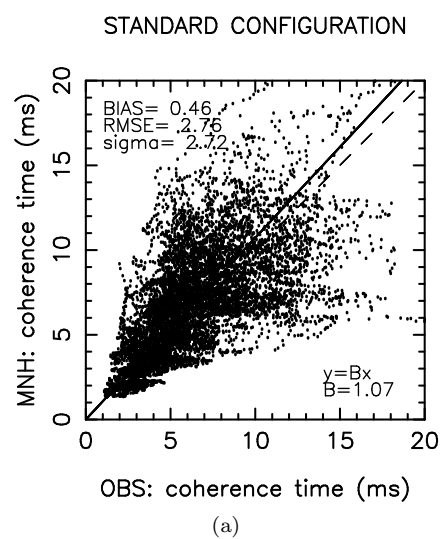


Figure 28: night3 - wavefront coherence time (ms): STANDARD CONFIGURATION

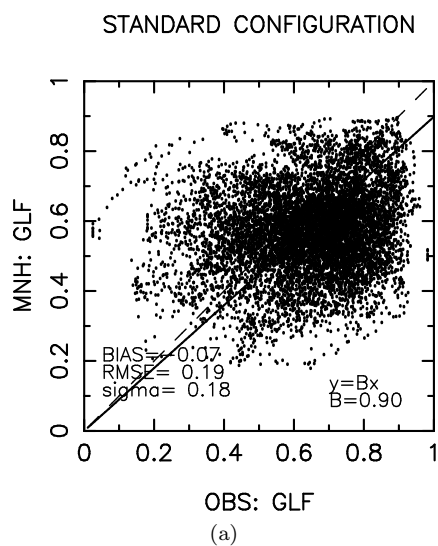


Figure 29: night3 - ground layer fraction (dimensionless): STANDARD CONFIGURATION

5.2 NIGHT3: Summarizing values of bias, RMSE, SD for long and short forecast time scales

VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS (m·s ⁻¹)	0.02	2.65	2.65	0.43	2.72	2.69
WD (degree)	-1.74	36.37	36.33	-8.05	27.01	25.78
RH (percent)	0.77	8.83	8.80	0.08	8.31	8.31
PWV (mm)	-0.52	1.76	1.68	-0.15	1.53	1.53
SEE (arcsec)	-0.04	0.23	0.23	-0.02	0.22	0.22
TAU (ms)	0.46	2.76	2.72	0.35	2.68	2.66
GLF (dimensionless)	-0.07	0.19	0.18	-0.06	0.18	0.17

Table 41: night3 - Statistics for variables in standard configuration: incremental month (i.e., since the beginning of service) *vs* last month (April-2024)

PARAMETER	STANDARD
POD1	57.2
POD2	50.4
POD3	75.3
PC	61.1
EBD	3.0

Table 42: night3 - PODs for wind speed (ms^{-1})

5.3 NIGHT3: Probability of detection (PODi) for long and short forecast time scale

FATE - Monthly Report

PARAMETER	STANDARD
POD1	88.9
POD2	48.0
POD3	77.6
POD4	27.4
PC	77.8
EBD	1.7

Table 43: night3 - PODs for wind direction (degree)

PARAMETER	STANDARD
POD1	34.3
POD2	56.5
POD3	76.7
PC	57.2
EBD	4.7

Table 44: night3 - PODs for relative humidity (percent)

PARAMETER	STANDARD
POD1	82.2
POD2	54.7
POD3	85.5
PC	76.3
EBD	1.0

Table 45: night3 - PODs for precipitable water vapor (mm)

PARAMETER	STANDARD
POD1	65.6
POD2	36.1
POD3	43.1
PC	48.9
EBD	7.5

Table 46: night3 - PODs for total seeing (arcsec) and accuray 0.0

PARAMETER	STANDARD
POD1	95.3
POD2	96.8
POD3	77.3
PC	93.4
EBD	5.6

Table 47: night3 - PODs for total seeing (arcsec) and accuray 0.24

FATE - Monthly Report

PARAMETER	STANDARD
POD1	63.9
POD2	41.0
POD3	87.5
PC	68.3
EBD	1.3

Table 48: night3 - PODs for wavefront coherence time (ms) and accuray 0.0

PARAMETER	STANDARD
POD1	84.5
POD2	68.8
POD3	90.9
PC	84.8
EBD	3.5

Table 49: night3 - PODs for wavefront coherence time (ms) and accuray 1.22

PARAMETER	STANDARD
POD1	50.5
POD2	44.5
POD3	19.4
PC	36.6
EBD	16.8

Table 50: night3 - PODs for ground layer fraction (dimensionless) and accuray 0.0

PARAMETER	STANDARD
POD1	88.6
POD2	89.3
POD3	44.4
PC	77.8
EBD	18.0

Table 51: night3 - PODs for ground layer fraction (dimensionless) and accuray 0.14

6 DAY3: Statistical analysis of forecast performances

6.1 DAY3: Scatter plots (bias, RMSE, SD) for long and short forecast time scale

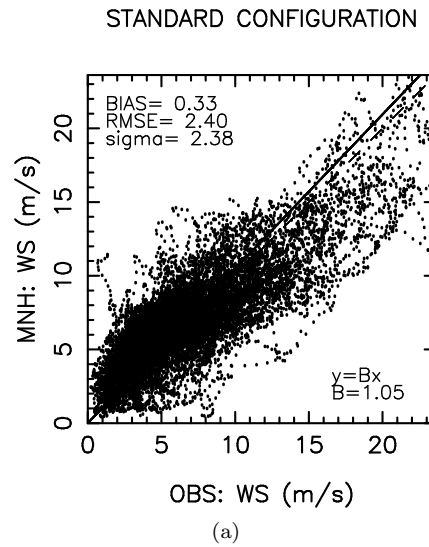


Figure 30: day3 - wind speed (ms^{-1}): STANDARD CONFIGURATION

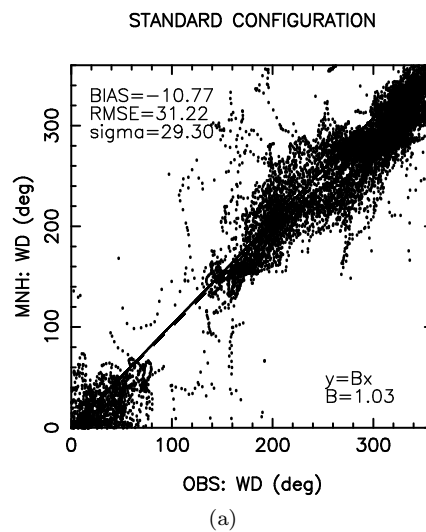


Figure 31: day3 - wind direction (degree): STANDARD CONFIGURATION

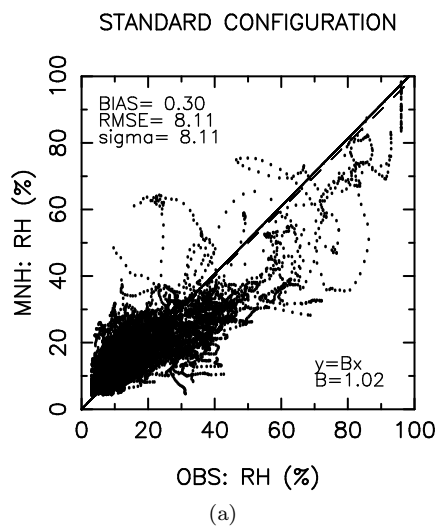


Figure 32: day3 - relative humidity (percent): STANDARD CONFIGURATION

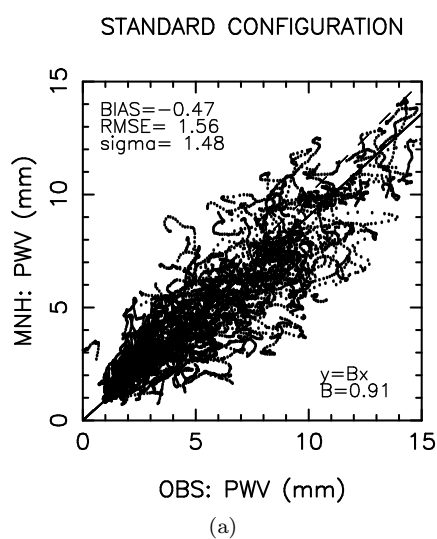


Figure 33: day3 - precipitable water vapor (mm): STANDARD CONFIGURATION

6.2 DAY3: Summarizing values of bias, RMSE, SD for long and short forecast time scales

VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS (m·s ⁻¹)	0.33	2.40	2.38	0.87	2.27	2.09
WD (degree)	-10.77	31.22	29.30	-13.30	25.97	22.30
RH (percent)	0.30	8.11	8.11	-0.70	7.76	7.72
PWV (mm)	-0.47	1.56	1.48	-0.35	1.58	1.54

Table 52: day3 - Statistics for variables in standard configuration: incremental month (i.e., since the begining of service) *vs* last month (April-2024)

PARAMETER	STANDARD
POD1	44.3
POD2	65.9
POD3	77.3
PC	64.1
EBD	1.5

Table 53: day3 - PODs for wind speed (ms^{-1})

6.3 DAY3: Probability of detection (PODi) for long and short forecast time scales

FATE - Monthly Report

PARAMETER	STANDARD
POD1	62.7
POD2	27.2
POD3	76.3
POD4	85.9
PC	70.8
EBD	0.7

Table 54: day3 - PODs for wind direction (degree)

PARAMETER	STANDARD
POD1	15.9
POD2	56.3
POD3	86.2
PC	63.1
EBD	2.0

Table 55: day3 - PODs for relative humidity (percent)

PARAMETER	STANDARD
POD1	63.2
POD2	66.9
POD3	86.0
PC	76.9
EBD	0.7

Table 56: day3 - PODs for precipitable water vapor (mm)

7 WP Statistical analysis of the forecast system failures and ESO observation availability

Table 57 reports the status of success rate in carrying out a complete forecast. Different potential sources of failure related to the operations are taken into account.

Each item in the first column starts with a label that indicates the origin of the 'potential' problem. Labels are: ECMWF, ESO, FATE. The success rate for the current month is calculated by the number of success cases divided by the number of days in a month. The success rate for the current incremental month is calculated by the number of success cases divided by the number of days since the start of the operations. The operations starting date is YYYYMMDD. The success rate is calculated using Eq.1:

$$(1 - M/E) \times 100 \quad (1)$$

where M is the number of failures and E is the number of full days. We mean with "full day" = 24h.

LEGEND:

- **ECMWF - ECMWF initialisation data transmission:** missing initialisation data coming from ECMWF. We consider a failure = 1 if data are not found in the main server 1a¹ (or 1b) in the related specific temporal windows on 24h. Server 1a is the main server located in ESO-Santiago (Chile), server 1b is the back-up server located in ESO-Garching (Germany).
- **FATE: networks:** failure on networks (main and back-up) in LaMMA. We consider a failure = 1 when both of them, the main network and the back-up network, do not allow the expected delivery on a full day.
- **ESO server INPUT problem:** server presenting bad operation. We have a failure = 1 when the server receiving the ECMWF data-set is not responding on a time scale of a full day and initialization data can not be downloaded in the temporal windows dedicated for carrying out the forecast for a full day. (Art. 7.1)
- **ESO - server OUPUT problem:** server presenting bad operation. We have a failure = 1 when the server receiving the delivery data-set from FATE is not responding and delivery data can not be uploaded (for night and day) neither on the main nor the back-up server within the time indicated in the technical specification. More precisely: two hours before the sunrise and the sunset (contract Art.7.1).
- **FATE - hydrodynamic computation failure:** missing forecasts. We have a failure = 1 when the missing forecasts are due to a hydrodynamic computation numerical instability that leads to the forecast failure.
- **FATE - delay or failure in providing deliverables:** missing deliveries. We have a failure = 1 when deliveries are not uploaded due to a FATE problem for a full forecast day (contract Art. 7.1).

The first five rows are independent items. The last row is the sum of the failures due to FATE. It is useful for the calculation of potential penalties.

Table 58 - Table 59 show the percentage of real-time observations retrieved from ESO archive daily in real-time calculated on the last month and on the whole period of the operational service starting from the starting date of operations up to the current month. The estimation is done for the night time (Table 58) and day time (Table 59).

The percentage is obtained through Eq.1 where M are the missing observations and E the expected observations. The expected observations are calculated considering the measurements ideally extended between the

¹See tech-note-20221019-001

sunset and sunrise (night) and sunrise and sunset (day) with resampled data on 5 minutes time scale. The choice of 5 minutes is the largest possible $\Delta(T)$ that allows to provide a forecast at short time scale (AR) as conceived in this project.

Automatic forecast system	Current month (success rate in %)	Incremental months (success rate in %)
ECMWF: ECMWF initialisation data transmission	100.00	99.00
FATE: principal network	100.00	98.00
ESO: server INPUT	100.00	100.00
ESO: server OUTPUT	100.00	98.00
FATE: hydrodynamic computation failure	100.0	100.0
FATE: delay on providing a forecast	100.00	100.00

Table 57: FATE: automatic forecasting system performance - statistics

Variable	Current month (success rate %)	Incremental months (success rate %)
SEE	79.700	86.100
TAU	82.000	85.400
GLF	83.200	87.200
WS	99.800	100.000
WD	99.800	100.000
RH	99.800	98.700
PWV	40.300	46.500

Table 58: night1 - real-time observations availability

Variable	Current month (success rate %)	Incremental months (success rate %)
WS	100.000	100.000
WD	100.000	100.000
RH	100.000	100.000
PWV	93.100	96.500

Table 59: day1 - real-time observations availability